

Straight-LOO mosaic on AWS — step-by-step

Goal: Rebuild the Run 1 3x3 data story so peer context on panels 1, 3, 7, 8, 9 uses straight LOO pond level (`pool_tb_ratio_mean_snr_fwd` / `z_pool_tb_ratio_mean_snr_fwd`) — basketball `poolq_loo` analog — instead of relative standing (`pool_minus_mean_snr_fwd` / `z_pool_minus_mean_snr_fwd`).

Unchanged panels: 2 (already shows T_j = straight LOO), 4, 5, 6.

Run from: 520 root (`Network_1P_shell/` or equivalent — folder with `big_dfs/`, `pipeline_config.py`, `520_pipeline_cox_working.ipynb`, `talent/re_entry/`).

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Step	Task	Status
0	Baseline upload + <code>active_at_eval_thru</code> BDP complete	☐
1	Choose Path A (mosaic-only) or Path B (z in feather)	☐
2	Edit <code>talent/re_entry/*.py</code> (+ optional manifest)	☐
3	Path B only: edit <code>pipeline_config_17_1.py</code> + Cells 10.5–11	☐
4	Probes → BDP pipeline → mosaic	☐
5	Compare PNGs to minus-mean baseline	☐

0. Prerequisites (do this first)

Complete the **baseline** upload from `AWS_UPLOAD_CHECKLIST.md` :

- `POOL_GROUPING_MODE = 'active_at_eval_thru'` in `pipeline_config.py`
- `POOL_EXCLUDE_PEER_TB_ZERO = False` (baseline; TB-zero sensitivity is a separate pass)
- Cells 0 → 11 already run; feather exists:

```
ls -lh big_dfs/df_pipeline_11_cox_analysis.feather
```

- Baseline minus-mean mosaic saved (for side-by-side):

```
cp talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png \
talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3_minus_mean.png
```

1. Choose your path

	Path A — mosaic-only	Path B — z in feather (recommended)
Notebook re-run	None (if columns already in feather)	Cell 10.5 → Cell 11
HERO x-axis	Raw <code>pool_tb_ratio_mean_snr_fwd</code> or on-the-fly z in Act II	<code>z_pool_tb_ratio_mean_snr_fwd</code> from Cell 10.5
Cox / CR plots	Unchanged (still minus-mean Run 1)	Run profile switches to straight LOO
Time on AWS	~5–15 min (plots only)	~30–90 min (+ standardize export)

Recommendation: Path B if Alex should see z-scored pond level (parity with current `z_pool_minus_mean` HERO). Path A for a quick visual before committing to a new Run profile.

2. Quick column probe (both paths)

From 520 root:

```
python3 - <<'PY'
import pandas as pd
p = "big_dfs/df_pipeline_11_cox_analysis.feather"
df = pd.read_feather(p)
cols = [
    "tb_ratio_fwd_snr",
    "pool_tb_ratio_mean_snr_fwd",
    "pool_minus_mean_snr_fwd",
    "z_tb_ratio_fwd_snr",
    "z_pool_minus_mean_snr_fwd",
    "z_pool_tb_ratio_mean_snr_fwd",
    "pool_size_snr_fwd",
]
for c in cols:
    ok = c in df.columns
    n = df[c].notna().sum() if ok else 0
    print(f" {c}: {'YES' if ok else 'MISSING'}  n={n:,}")
PY
```

- `pool_tb_ratio_mean_snr_fwd` should be **YES** after any successful Cell 5+ run.
- `z_pool_tb_ratio_mean_snr_fwd` is **MISSING** until Path B (Cell 10.5 with updated standardize list).

Pool-size sanity (unchanged by LOO metric choice):

```
python -u talent/re_entry/army_pool_size_probe.py
```

3. Path B — notebook + Run profile (z column in feather)

3a. Confirm Cell 0 loads Run 1 override

Open `520_pipeline_cox_working.ipynb` → Cell 0 (session/import cell).

Cell 0 line	Check
L57 <code>pip_config_file = '...'</code>	Must resolve to <code>pipeline_config_17_1</code> on AWS (not <code>pipeline_config_div_name</code> unless that file mirrors Run 1).
L166 (after run)	Prints <code>STANDARDIZE_CONFIG['cols']</code> — note current list.

Re-run Cell 0 after any edit to `pip_config_file`.

3b. Edit `pipeline_config_17_1.py` (520 root)

Replace **pool minus mean** with **straight LOO mean** in the Run 1 profile:

Line	From	To
34 <code>tv_vars</code>	<code>[TB_RATIO_SNR_COL, POOL_MINUS_MEAN_SNR_COL]</code>	<code>[TB_RATIO_SNR_COL, POOL_TB_RATIO_MEAN_SNR_COL]</code>
35 <code>quadratic_bases</code>	<code>[TB_RATIO_SNR_COL, POOL_MINUS_MEAN_SNR_COL]</code>	<code>[TB_RATIO_SNR_COL, POOL_TB_RATIO_MEAN_SNR_COL]</code>
36 <code>combined_effect_base</code>	<code>POOL_MINUS_MEAN_SNR_COL</code>	<code>POOL_TB_RATIO_MEAN_SNR_COL</code>
40 <code>var_y_label</code>	<code>'Pool minus mean'</code>	<code>'Pool LOO mean ($\hat{\tau}_j$)'</code>

CR plot block (lines 77–81): auto-updates on next Cell 0 reload because `_cr_var_pool` uses `POOL_MINUS_MEAN_SNR_COL` today — after the swap it becomes `z_pool_tb_ratio_mean_snr_fwd` when `RUN_SCALE='z'`.

Optional semantic alias in `pipeline_config.py` (lines 258–259):

```
POOL_CONTEXT_SNR_COL = POOL_TB_RATIO_MEAN_SNR_COL # was POOL_MINUS_MEAN_SNR_COL
```

Not required for the mosaic; documents the Run fork.

3c. Notebook run order (Path B)

Order	Notebook cell	What it does
1	Cell 0	Reload configs; verify <code>STANDARDIZE_CONFIG['cols']</code> includes <code>pool_tb_ratio_mean_snr_fwd</code>
2	Cell 10.5	Creates <code>z_pool_tb_ratio_mean_snr_fwd</code> (+ interaction/quadratic chain)
3	Cell 11	Re-exports <code>big_dfs/df_pipeline_11_cox_analysis.feather</code>

Do not re-run Cell 5 unless you changed pool grain or `PPOOL_EXCLUDE_*` — straight LOO mean already exists in the feather from Cell 5.
Re-run the column probe (§2); expect `z_pool_tb_ratio_mean_snr_fwd: YES`.

4. Edit plotting scripts (both paths)

All paths below: edit files under `talent/re_entry/` on AWS (same paths as Mac).

4a. `army_basic_plots.py` — Panel 3

Line	Change
42	<code>COL_L00 = "pool_tb_ratio_mean_snr_fwd"</code> (was <code>pool_minus_mean_snr_fwd</code>)
631	x-label → <code>pool_tb_ratio_mean_snr_fwd (LOO pond level)</code>
633	title → <code>Peer pool quality (straight LOO mean)</code>
646	x-label → <code>pool_tb_ratio_mean_snr_fwd</code>

Note: Output filename stays `ARMY_BDP_pool_minus_mean_loo_run1.png` (stem at line 653) — content changes; manifest path unchanged.
Optional: change stem to `_pool_tb_ratio_mean_loo_` and update manifest panel 3 path.

4b. `army_act2_probes.py` — Panels 7 & 8

Line	Change
55	<code>COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"</code>
59	In <code>OER_CHECK_COLS</code> , keep both raw columns; order does not matter

Fallback behavior (lines 247–250): if z column missing, Act II z-scores `COL_L00` within the filtered panel. So Path A works when 42 in `army_basic_plots.py` is synced and you set `COL_L00` import — Act II imports `COL_L00` from `army_basic_plots`, so 4a must be done first.

4c. `army_hero_slide_plot.py` — Panel 9

Line	Path A	Path B
44 <code>plot_var</code>	<code>"pool_tb_ratio_mean_snr_fwd"</code>	<code>"z_pool_tb_ratio_mean_snr_fwd"</code>

Or pass at runtime (no default edit):

```
./talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_tb_ratio_mean_snr_fwd
```

Expected output: `talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_tb_ratio_mean_snr_fwd_run1.png`

5. Manifest (panels 1 & 9 + footer)

Option 1 — overwrite default manifest (`manifests/army_run1_3x3_manifest.json`):

Field	New value
<code>title</code>	<code>... straight LOO pond (↑j) ...</code>
<code>subtitle</code> / <code>footer</code>	Replace <code>z_pool_minus_mean</code> → <code>z_pool_tb_ratio_mean_snr_fwd</code>
Panel 1 line Peer X:	<code>z_pool_tb_ratio_mean_snr_fwd (straight LOO HERO)</code>
Panel 1 Reigning tag:	<code>ew8_z_pool_tb_ratio_mean_snr_fwd_run1</code>
Panel 9 <code>path</code>	<code>.../ARMY_HERO_ew8_z_pool_tb_ratio_mean_snr_fwd_run1.png</code>
<code>verdict.hero_reigning</code>	<code>ew8_z_pool_tb_ratio_mean_snr_fwd_run1</code>
<code>verdict.peer_pool</code>	<code>senior-rater straight LOO mean (forward SNR)</code>

Option 2 — side-by-side deck (recommended): Copy manifest to `manifests/army_run1_3x3_straight_loo_manifest.json`, edit copy only, build with:

```
./talent/re_entry/build_army_data_story.py \  
--manifest talent/re_entry/manifests/army_run1_3x3_straight_loo_manifest.json
```

Output: set `"output_png"` in that manifest (e.g. `ARMY_DATA_STORY_run1_3x3_straight_loo.png`).
Panel 3 path: unchanged unless you renamed the BDP stem in §4a.

6. Run order on AWS

From 520 root, after script edits (and Path B feather refresh):

```
chmod +x talent/re_entry/*.py talent/re_entry/*.sh

# 1 - pool probe (informational; same feather)
python -u talent/re_entry/army_pool_size_probe.py

# 2 - BDP panels 2-6 + panel 3 + Act II + HERO
bash talent/re_entry/run_army_bdp_pipeline.sh
```

Or stepwise (easier to debug):

```
./talent/re_entry/army_basic_plots.py --all
./talent/re_entry/army_act2_probes.py --plot all_probes
./talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_tb_ratio_mean_snr_fwd # Path B
# ./talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd # Path A
./talent/re_entry/build_army_data_story.py
# add --manifest ... if using straight_loo manifest copy
```

Do not need for this mosaic fork: Cell 12 Cox refit (unless you want CR/Cox slides aligned with Path B profile).

7. Success checklist

Artifact	Straight LOO expected
Panel 3 PNG	Histogram centered on positive pond TB levels (not mean $\approx$ 0 like minus-mean)
Panel 7	ARMY_CCT_promotion_rate_pool_loo_run1_z1_2_q8.png — bins use pond z, not minus-mean z
Panel 8	ARMY_ELITE_pond_loo_pw3p5_run1_top20.png
Panel 9	ARMY_HERO_ew8_z_pool_tb_ratio_mean_snr_fwd_run1.png
Mosaic	ARMY_DATA_STORY_run1_3x3_straight_loo.png (or overwritten default)

Spot-check panel 9 sidecar JSON:

```
cat talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_tb_ratio_mean_snr_fwd_run1.json | head -20
```

Confirm "plot_var": "z_pool_tb_ratio_mean_snr_fwd".

8. What each panel uses (reference)

Panel	Variable after switch
1	Text only
2	tb_ratio_fwd_snr vs pool_tb_ratio_mean_snr_fwd (no change)
3	pool_tb_ratio_mean_snr_fwd
4	tb_ratio_fwd_snr only
5	Pool overlap on z_tb_ratio_fwd_snr + membership
6	pool_size_snr_fwd
7-8	z_pool_tb_ratio_mean_snr_fwd (or on-the-fly z of raw LOO mean)
9	z_pool_tb_ratio_mean_snr_fwd (or raw)

9. Rollback

1. Restore COL_LOO / COL_LOO_Z / plot_var in the three .py files.
2. Restore pipeline_config_17_1.py tv_vars to POOL_MINUS_MEAN_SNR_COL if Path B was applied; re-run Cell 10.5 + 11.
3. Re-run run_army_bdp_pipeline.sh with original manifest.
4. Baseline PNG: ARMY_DATA_STORY_run1_3x3_minus_mean.png (\$0).

10. Notebook cell index map (Mac canonical 520_pipeline_cox_working.ipynb)

Display label	Notebook cell index	Key lines
Cell 0 (imports / config)	2	L57 <code>pip_config_file</code> ; L175 pool toggles
CELL 5 pool means	20	L15–34 <code>add_pool_means_and_sizes(...)</code>
CELL 10.5 standardize	48	L9–11 <code>STANDARDIZE_CONFIG</code>
CELL 11 Cox export	53	L95–98 reads standardize cols

Cell 5 already writes `pool_tb_ratio_mean_*` and `pool_minus_mean_*` together; switching the mosaic does **not** require re-running Cell 5.

Thread log

- 2026-09-21 — Initial guide: Path A/B, file line refs, AWS run order, probes, manifest fork.